NUTRIVIET: SUMMARY OF CHANGES (PA3-2026 SUBMISSION)

Course: CS300 - CSC13002 - Introduction to Software Engineering
Project Assignment: PA3-2026
Group: Group 07 - NutriViet Team
Submission Date: 14/11/2026

OVERVIEW

This `Changes.md` document provides a comprehensive, traceable record of all modifications, extensions, and refinements implemented in the NutriViet project artifacts for the **Project Assignment 3 (PA3-2026)** milestone compared to the previous PA2-2026 baseline. In accordance with PA3 guidelines, changes are systematically grouped by document section.

SECTION A - CHANGES IN REVISED PROJECT PLAN (`DOCS/MANAGEMENT/PROJECT-PLAN.MD`)

Performed by: Ngo Tien Binh | Reviewed by: Le Tat Nam | Edited by: Truong Gia Linh

#	Section / Component	Type of Change	Description & Academic Rationale
1	**Header & Metadata**	*Updated*	Promoted document status from PA2-2026 to **PA3-2026 (2nd Submission)**.
2	**Section 3.1 Team Structure**	*Refined*	Clarified Luong Hoang Nam's responsibilities to explicitly encompass high-fidelity interactive UI prototyping and screenshot evidence generation for PA3 Section D.
3	**Section 4.1 5-Sprint Roadmap**	*Updated*	Marked Sprint 2 (PA2) as **Completed** (18/10/2026 – 31/10/2026). Transitioned Sprint 3 (PA3: 01/11/2026 – 14/11/2026) to **Active Sprint**, detailing core objectives: Use-Case Modeling (Section C), Use-Case Specifications & UI Prototypes (Section D), Spec Kit End-to-End Implementation of 1 Functional Group (Section E), and Scrum/AI reporting (Section F).
4	**Section 4.2 Sprint 2 Archive**	*Restructured*	Formally archived the 14 completed tasks of Sprint 2 (`TASK-01` to `TASK-14`) as historical project evidence.
5	**Section 4.3 Sprint 3 Active Tasks**	*Added (Major)*	Created a granular 14-task work breakdown structure for Sprint 3 (`TASK-15` through `TASK-28`). In strict accordance with course rules, each task has **exactly 1 designated performer**, **1 designated reviewer**, start/due dates, and synchronized progress status (*In Progress* / *To Do*). Tasks directly map to PA3 grading sections A through F.
6	**Section 4.4 Later Sprints**	*Refined*	Refined planned activities for Sprint 4 (core system integration & test coverage $\ge 80%$) and Sprint 5 (crowdsourcing, admin moderation, and live demonstration).

SECTION B - CHANGES IN DETAILED VISION DOCUMENT (`DOCS/REQUIREMENTS/VISION.MD`)

Performed by: Le Tat Nam | Reviewed by: Truong Gia Linh | Edited by: Nguyen Hoang Phuc Thinh

#	Section / Component	Type of Change	Description & Academic Rationale
1	**Header & References**	*Updated*	Updated document header to **PA3-2026 (2nd Submission)**. Added formal academic references to **WCAG 2.1 Guidelines** for accessibility standards.
2	**Section 3.3 User Environment**	*Expanded (Major)*	Substantially expanded the user environment specification into 4 exhaustive dimensions: 1. *Physical Retail Environments:* Illumination variations (150 to 600 lux), cellophane/tin packaging reflections, packaging curvature, and one-handed mobile ergonomic constraints in grocery store aisles. 2. *Mobile Hardware Diversity:* Android fragmentation (API 26 to 35), camera sensor specs (8MP fixed-focus to 50MP OIS), and RAM capacities (2GB entry-level to 12GB flagship). 3. *Network Infrastructure:* Supermarket basement connectivity drops (cellular packet loss down to 0 kbps) and the necessity of autonomous local offline fallback. 4. *Administrative Desktop Environment:* Desktop browser requirements (Chrome 120+, Firefox, Safari, Edge on 1080p+ viewports).
3	**Section 3.5 Alternatives & Competition**	*Expanded (Major)*	Formulated a comprehensive **8-dimension comparative evaluation matrix** benchmarking NutriViet against 4 primary alternatives: Yuko App (EU), Soosee App (NL), Open Food Facts, and MyFitnessPal. Added MyFitnessPal to the positioning chart to evaluate general dietary tracking apps.
4	**Section 5.1 Feature Descriptions**	*Refined (Major)*	Decomposed each of the 8 Functional Groups (FG1 through FG8) into concrete sub-capabilities with explicit 1:1 traceability tags mapping directly to Use Cases `UC01` through `UC18` in Section C & D.
5	**Section 6 Non-Functional Requirements**	*Refined & Added*	Appended formal **Verification Methods** (automated benchmarks, packet inspection, Locust stress test, Firebase Test Lab) to all existing performance, scalability, security, platform, and reliability requirements to ensure testability.
6	**Section 6.6 Usability & Accessibility**	*Added (New)*	Added `NFR-USAB-01` mandating WCAG 2.1 AA dual-coding for the tri-tier safety alerts: every color indicator (RED/YELLOW/GREEN) must be paired with distinct geometric glyphs (`[!X]`, `[!?]`, `[OK]`) and enforce a minimum 4.5:1 contrast ratio to protect color-blind consumers.

SECTION C - CHANGES IN USE-CASE MODEL (`DOCS/REQUIREMENTS/USE-CASES.MD`)

Performed by: Luong Hoang Nam | Reviewed by: Ngo Tien Binh | Edited by: Le Tat Nam

#	Section / Component	Type of Change	Description & Academic Rationale
1	**System-Level Mermaid Diagram**	*Added (New)*	Created a comprehensive system-level Use-Case diagram visualizing all 18 use cases across FG1–FG8 mapped to primary actors (Consumer, System Administrator specializing User) and secondary technical actors (Google ML Kit, FoodOn Semantic Ontology Engine).
2	**Subsystem Mermaid Diagrams & Relationship Analysis**	*Added (New)*	Constructed 4 detailed subsystem diagrams and relationship analysis table illustrating `<<include>>`, `<<extend>>` (with explicit conditions), and generalization (actor & use case variants) across core functional workflows.

#	Section / Component	Type of Change	Description & Academic Rationale
3	**Use Case Categorization Table**	*Added (New)*	Mapped all 18 use cases directly to Functional Groups FG1 through FG8 with formal IDs (`UC01` to `UC18`).

SECTION D - CHANGES IN USE-CASE SPECIFICATIONS & PROTOTYPES (`DOCS/REQUIREMENTS/USE-CASES.MD`)

Performed by: Truong Gia Linh & Nguyen Hoang Phuc Thinh | Reviewed by: Luong Hoang Nam

#	Section / Component	Type of Change	Description & Academic Rationale
1	**18 Detailed Use-Case Specs**	*Added (Major)*	Formulated formal use-case specifications for all 18 use cases featuring Preconditions, Basic Flow, Alternative Flows (covering exception states, bad camera focus, network failure), and Postconditions.
2	**High-Fidelity UI Prototypes**	*Added (Major)*	Engineered and embedded **17 high-resolution prototype screens** (`assets/prototypes/`) illustrating consumer interfaces, hazard alert dual-coding, scanning viewfinders, diary logging, crowdsource photo upload, and admin moderation.

SECTION E - SPEC KIT FULL-STACK IMPLEMENTATION (`SPECS/` & `SRC/`)

Performed by: Le Tat Nam, Luong Hoang Nam, Ngo Tien Binh | Reviewed by: Truong Gia Linh

#	Section / Component	Type of Change	Description & Academic Rationale
1	**Spec Kit Lifecycle Artifacts**	*Added (Major)*	Authored complete Spec-Driven Development cycle in `specs/001-authentication/` and `specs/002-allergy-profile/`: `spec.md`, `plan.md`, `data-model.md`, `openapi.json`, `tasks.md`, `research.md`, and `requirements.md` (22/22 gates passed).
2	**Backend Microservice**	*Added (Major)*	Built production Python WSGI microservice (`wsgi_app.py`) deployed live on PythonAnywhere (`ngobinh2006.pythonanywhere.com`) with PBKDF2 HMAC-SHA256 crypto, password change, body physical metrics (BMI/TDEE), and SQLite relational persistence.
3	**Mobile Client Application**	*Added (Major)*	Developed responsive Flutter mobile application (`src/mobile/`) with zero-prefill login, email activation gate, dedicated password change screen, BMI/TDEE physical metrics, modern 5-tab taskbar with glowing center QUÉT button, and crowdsourced product photo upload.
4	**Video Demonstration**	*Added (Major)*	Recorded and published full-stack video demonstration on YouTube accompanied by structured bilingual narration script (`video-demo-script.md`).

SECTION F - CHANGES IN WEEKLY SCRUM & AI LOGS (`DOCS/MANAGEMENT/WEEKLY-REPORTS.MD`)

Performed by: Truong Gia Linh | Reviewed by: Ngo Tien Binh

#	Section / Component	Type of Change	Description & Academic Rationale
1	**Scrum Meeting Minutes**	*Added (New)*	Documented 4 sprint ceremonies across Sprint 3 (01/11 – 14/11/2026): Sprint 3 Planning (02/11), Mid-Sprint Standup (08/11), Mid-Sprint Progress & Spec Kit Checkpoint (12/11), and Sprint Review & Retrospective (14/11).
2	**Trello Progress Tracking**	*Updated*	Embedded 3 sequential Trello board progression checkpoints across Sprint 3: Mid-sprint (08/11: 3 Done, 6 In Progress, 5 To Do - Figure 1.1), Spec Kit Checkpoint (12/11: 9 Done, 5 In Progress, 0 To Do - Figure 1.2), and Final Sprint Delivery (14/11: 14/14 Done (100%), 0 In Progress, 0 To Do - Figure 1.3) verifying rigorous 1:1 member task assignments and 100% completion.
3	**Generative AI Disclosures**	*Added (New)*	Compiled comprehensive AI interaction log with prompts, outputs, and human verification rationale alongside a ratified Academic Integrity Statement.

SUMMARY OF COMPLETED STAGE 1 & STAGE 2 DELIVERABLES

- ☑ **Section A:** Revised Project Plan (project-plan.md) + `Changes.md`
- ☑ **Section B:** Detailed Vision Document (vision.md) + `Changes.md`
- ☑ **Section C:** Use-Case Model with Mermaid Diagrams (use-cases.md)
- ☑ **Section D:** Use-Case Specifications & High-Fidelity UI Prototypes (use-cases.md)
- ☑ **Section E:** Full-Stack Implementation using Spec Kit - FG1 Authentication & FG2 Profile (specs/)
- ☑ **Section F:** Weekly Scrum Reports & Generative AI Tool Disclosures (weekly-reports.md)